A Credible-Subpopulation Approach to Parallel Trends

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July 20, 2026

Abstract

Difference-in-differences (DiD) practitioners routinely confront the possibility that the parallel trends assumption fails for some treated cohorts but not others. The dominant responses keep the average treatment effect on the treated (ATT) as the target: heterogeneity-robust estimators point-identify it under parallel trends, while sensitivity methods such as HonestDiD report a set for it under bounded violations. Both inherit the problem that the ATT may be either biased or only weakly set-identified when some cohorts violate parallel trends. We propose a different response: change the estimand. We define a *credible-subpopulation local ATT* (LATT), the treatment effect for the subpopulation of cohorts whose parallel trends assumption is credible, and show it is point-identified precisely when the full ATT is not. The procedure is a reweighting of standard group-time effects toward credible cohorts, combined with honest sensitivity bounds on the residual violation of the selected set. We are explicit about the method's scope: its advantage over the ATT grows with how informative pre-trends are about post-treatment violations and vanishes when they are uninformative, where only economic context can carry the identifying assumption. A reduced-form and a panel simulation study confirm that the LATT recovers a clean effect where the ATT estimator is biased, that the sensitivity bounds restore honest coverage where point estimates collapse, and that a relative-magnitudes restriction is self-undermining under uninformative pre-trends while a smoothness restriction is not.

1 Introduction

Difference-in-differences is among the most widely used research designs in empirical economics, and a large recent literature has clarified both its assumptions and its failure modes (Roth et al., 2023). The design's credibility rests on the parallel trends assumption: absent treatment, the average outcomes of treated and comparison units would have evolved in parallel. In practice researchers are rarely certain this holds, and it has become standard to assess it by testing for pre-treatment differences in trends. As Roth et al. (2023) synthesize, this pre-testing practice is fragile: pre-trends tests are often underpowered against economically meaningful violations (Kahn-Lang and Lang, 2020; Roth, 2022), conditioning on passing them induces a pre-test selection bias (Roth, 2022), and even a clean pre-period does not guarantee a clean post-period (Kahn-Lang and Lang, 2020).

Two strands of methodology respond to this concern, and both retain the ATT as the estimand. The first develops heterogeneity-robust estimators for staggered adoption (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; Borusyak et al., 2024; de Chaisemartin and D'Haultfœuille, 2020; Goodman-Bacon, 2021), which point-identify interpretable aggregations of group-time effects under a suitable parallel trends assumption but remain biased if that assumption fails. The second, exemplified by Rambachan and Roth (2023), abandons exact parallel trends and instead bounds how different post-treatment violations can be from the observed pre-trends, reporting a uniformly valid confidence *set* for the ATT. This is a substantial advance, but it inherits a structural limitation: when parallel trends fails for some cohorts, the ATT—an average over *all* treated cohorts, including the offending ones—is precisely the object that is hard to recover. The heterogeneity-robust point estimate is then biased, and the HonestDiD set can be wide and uninformative, because it must accommodate the worst offenders.

This paper takes a different route, familiar from other corners of causal inference: when the population target is not credibly identified, retreat to a subpopulation where it is. This is the logic of the local average treatment effect (Imbens and Angrist, 1994), of the optimal-subpopulation average treatment effect under limited overlap (Crump et al., 2009), and of overlap weighting (Li et al., 2018). In each case one trades the original estimand for a credibly-identified effect on a selected subpopulation, and defends the trade by characterizing whom the subpopulation comprises. We develop the staggered-DiD instance of this principle. We define the credible-subpopulation LATT—the weighted average of group-time effects restricted to cohorts whose parallel trends assumption is credible—and show that it is point-identified even when the full ATT is not, because the violations of the excluded cohorts do not enter it. Estimation is a transparent reweighting of the Callaway and Sant'Anna (2021) group-time effects; the credibility of a cohort is assessed from its pre-trends, so that the reweighting is a data-dependent selection, and we treat inference on the selected set with the sensitivity machinery of Rambachan and Roth (2023).

Our contribution is therefore primarily one of framing and estimand choice rather than of new estimation or inference technology, and we are deliberate about this. The method occupies a distinct point on a credibility–precision–scope frontier: relative to a heterogeneity-robust point estimate of the ATT it gives up scope (it targets a subpopulation) to buy back credibility; relative to a HonestDiD set for the ATT it gives up nothing in honesty but recovers a sharper, point-identified answer to a narrower question. We are equally explicit about the method's limits. Its advantage is governed by how informative pre-trends are about post-treatment violations; when they are uninformative, selecting on them buys nothing, and the identifying assumption must be carried by economic context rather than by the data. We show, moreover, that the natural relative-magnitudes sensitivity restriction is *self-undermining* in exactly this regime—it anchors its bounds to the observed pre-trends and so reports false precision when those pre-trends are flat—whereas a smoothness restriction, which keys off structure rather than magnitude, is not.

The remainder of the paper is organized as follows. Section 2 fixes the staggered-DiD setup and the decomposition of event-study coefficients into a causal and a differential-trend component. Section 3 defines the credible-subpopulation LATT, states its identification and the properties of its estimator, and gives our recommended sensitivity-based inference together with an honest account of what is provable and what is not. Section 4 reports the simulation study. Section 5 concludes.

2 Setup

We adopt the staggered adoption framework of Callaway and Sant'Anna (2021). There are periods $t = 1, \ldots, T$ and units i indexed by their first treatment date $G_i \in \{2, \ldots, T\} \cup \{\infty\}$, where $G_i = \infty$ denotes never-treated. Let $Y_{it}(g)$ denote the potential outcome in period t if unit i is first treated in period g, and $Y_{it}(\infty)$ the never-treated potential outcome. The building-block causal parameter is the group-time average treatment effect on the treated,

$$\mathrm{ATT}(g,t) = \mathbb{E}[\, Y_{it}(g) - Y_{it}(\infty) \mid G_i = g\,]\,. \tag{1}$$

Summary parameters are weighted aggregations $\theta = \sum_g \sum_t w(g,t)\,\mathrm{ATT}(g,t)$ with researcher-chosen weights, of which the overall ATT and the event-study path are special cases.

It is convenient to work with the event-study representation and the decomposition of Rambachan and Roth (2023). Let $\hat{\beta} = (\hat{\beta}'_{\mathrm{pre}}, \hat{\beta}'_{\mathrm{post}})'$ be an asymptotically normal vector of event-study coefficients estimated by any of the heterogeneity-robust procedures above, with $\sqrt{n}(\hat{\beta} - \beta) \to \mathcal{N}(0, \Sigma)$. The estimand decomposes as

$$\beta = \tau + \delta, \qquad \tau_{\mathrm{pre}} = 0, \tag{2}$$

where τ collects the causal effects (zero before treatment, by no anticipation) and δ is the differential trend between treated and comparison groups that would have occurred absent treatment. Parallel trends is the restriction $\delta_{\mathrm{post}} = 0$, under which $\beta_{\mathrm{post}} = \tau_{\mathrm{post}}$; a pre-trends test is a test of $\delta_{\mathrm{pre}} = 0$.

We will require the analogous objects at the cohort level. Write δ_g for the differential trend of cohort g and say that parallel trends *holds for g* if $\delta_{g,\mathrm{post}} = 0$. A key feature of applications is that this may hold for some cohorts and fail for others: a policy adopted by cohort g may be as good as randomly timed while another cohort's adoption coincides with a confounding shock.

Finally, we distinguish two regimes for how cohorts are judged credible, because they carry different inferential guarantees. Under *ex-ante* (or independent) selection, credibility is decided from pre-determined covariates, institutional knowledge, or an independent data split, so that the selected set is statistically independent of the estimation errors in $\hat{\beta}$. Under *data-driven* selection, credibility is assessed from the realized pre-trends $\hat{\beta}_{\mathrm{pre}}$, so that the selected set depends on the estimation sample. The distinction is immaterial for the estimand and its identification, but central for inference, and we treat it explicitly in Section 3.

3 The credible-subpopulation LATT: approach and recommendations

3.1 Estimand and identification

Let S denote a set of cohorts judged to have credible parallel trends, produced by a selection rule R that we specify below. We define the target as the treatment effect for that subpopulation.

Definition 1 (Credible-subpopulation LATT)**.** *For cohort weights $w_g > 0$ (e.g. proportional to cohort size), the credible-subpopulation local ATT is*

$$\theta_S \;=\; \frac{\sum_{g\in S} w_g \,\mathrm{ATT}(g,\cdot)}{\sum_{g\in S} w_g}, \tag{3}$$

where $\mathrm{ATT}(g,\cdot)$ *is any within-cohort aggregation of interest (a fixed event-time, or an average over post-treatment periods).*

The value of the target is that it is identified under a weaker condition than the ATT.

Proposition 1 (Identification)**.** *If parallel trends holds for every $g \in S$, i.e. $\delta_{g,\mathrm{post}} = 0$ for all $g \in S$, then θ_S is point-identified by the reweighted group-time effects, $\theta_S = \big(\sum_{g\in S} w_g\, \beta_{g,\mathrm{post}}\big)/\sum_{g\in S} w_g$, and this holds irrespective of whether $\delta_{g,\mathrm{post}} \neq 0$ for cohorts $g \notin S$.*

Proof. For $g \in S$, $\beta_{g,\mathrm{post}} = \tau_{g,\mathrm{post}} + \delta_{g,\mathrm{post}} = \tau_{g,\mathrm{post}}$ by the hypothesis and (2). The reweighted sum over S therefore equals the corresponding weighted average of causal effects, which is θ_S by definition. Cohorts outside S receive zero weight, so their differential trends do not enter. □

Proposition 1 is elementary, but it is the paper's central point: by *changing the target* to the credible subpopulation, one converts a partially-identified problem (the ATT, when some cohorts violate parallel trends) into a point-identified one, at the cost of answering a narrower question. This is exactly the trade made by Imbens and Angrist (1994), Crump et al. (2009), and Li et al. (2018) in other settings, and it carries the same interpretive obligation: one must characterize the subpopulation S that the estimand describes.

3.2 Estimation

The estimator $\hat{\theta}_S$ replaces population objects by their Callaway and Sant'Anna (2021) sample analogues and reweights over the selected cohorts. Its large-sample behavior is standard under ex-ante selection.

Proposition 2 (Consistency and efficiency)**.** *Under ex-ante selection and the regularity conditions of Callaway and Sant'Anna (2021), $\hat{\theta}_S$ is consistent for θ_S. If parallel trends holds for all cohorts, $\hat{\theta}_S$ is consistent for the ATT, with an efficiency loss relative to estimators that use all cohorts.*

Under data-driven selection the selected set is random and correlated with the estimation errors, which introduces a finite-sample pre-test bias of the kind analyzed by Roth (2022).

Proposition 3 (Pre-test bias)**.** *Under data-driven selection, $\hat{\theta}_S$ carries a finite-sample bias that vanishes as the per-cohort information grows and the selection rule becomes consistent for the credible set.*

We emphasize, consistent with Proposition 3 and with our simulations below, that the honest claim is consistency, not finite-sample unbiasedness: the data-driven estimator is close to, but not exactly on, the target in small samples, and the gap closes with sample size.

3.3 Honest inference on the selected set

Selecting credible cohorts does not by itself deliver parallel trends: a cohort with flat pre-trends may still have a nonzero post-treatment differential trend, and the selection removes gross offenders but not subtle residual violations. We therefore recommend accompanying the point estimate with the sensitivity analysis of Rambachan and Roth (2023), applied to the *selected-cohort aggregate* event study. Concretely, one imposes a restriction $\delta_{S,\text{post}} \in \Delta$ on the residual differential trend of the selected set and reports the implied confidence set for θ_S. Among the restrictions of Rambachan and Roth (2023), the smoothness class $\Delta^{SD}(M) = \{\delta : |\delta_{t+1} - 2\delta_t + \delta_{t-1}| \leq M \ \forall t\}$ admits the fixed-length confidence intervals (FLCIs) of Armstrong and Kolesár (2018), which are near-optimal for convex centrosymmetric restrictions and which we adopt.

The validity of this step depends on the selection regime. Under ex-ante selection the selected set is independent of $\hat{\beta}$, so the uniform validity results of Rambachan and Roth (2023) apply verbatim to the selected-cohort event study, and no new inference theory is required.

Under data-driven selection the situation is more delicate, and it is worth being precise about what is and is not implied, since the relevant impossibility result is easily overread. Leeb and Pötscher (2005) show that the distribution of a post-model-selection estimator cannot be estimated uniformly consistently, so that naive post-selection confidence intervals lack *uniform* asymptotic validity: formally, the supremum of the coverage error over the parameter space does not vanish. This is an existence statement about the worst case, not a claim that coverage fails at a typical parameter value. The obstruction is concentrated in *local-to-threshold* configurations, in which the parameter governing selection drifts toward the decision boundary of the selection rule at the sampling rate; in that region the selection probability is asymptotically interior, the post-selection distribution is non-normal and depends on the drift, and no plug-in can track it. Away from the boundary, at a fixed data-generating process in which cohorts are either clearly credible or clearly not, selection becomes asymptotically deterministic and post-selection inference behaves conventionally.

This resolves what would otherwise appear to be a tension with our simulation evidence. The simulations of Section 4 are pointwise experiments—they fix a data-generating process and vary the sampling noise—and in that regime we find no material coverage distortion, with full-data and split implementations crossing nominal coverage at the same sensitivity bound. But the simulations also exhibit the Leeb and Pötscher (2005) phenomenon exactly where the theory locates it. When cohorts are placed at the selection threshold, naive coverage does deteriorate relative to splitting;

and in the panel simulation the estimated standard error of the LATT falls materially short of its Monte Carlo counterpart at *intermediate* informativeness, which is precisely the regime in which selection is genuinely stochastic. The honest summary is therefore not that data-dependent selection is innocuous, but that its distortion arises where theory predicts and is small in magnitude in the configurations we examine.

Two features of the present setting further mute the effect. The estimand is an aggregate over many cohorts, so the distortion contributed by a borderline cohort is diluted by averaging over the cohorts whose selection is not in doubt, in contrast to the leading cases of the model-selection literature in which the inclusion of the parameter of interest is itself the decision. And the interval we recommend has width governed principally by the identification allowance Δ rather than by sampling variability, so a modest selection-induced perturbation of the sampling distribution is absorbed by an interval that is already wide for identification reasons.

We therefore do not read Leeb and Pötscher (2005) as an argument against the data-driven procedure, but as a statement about the kind of guarantee it can carry: pointwise validity in regimes away from the selection boundary, established here by simulation, rather than a worst-case uniform guarantee. A researcher who requires the latter can obtain it cheaply by selecting on pre-determined information, by splitting the sample, or by randomized (data-carving) selection in the sense of Fithian et al. (2014), each of which restores the independence that the ex-ante case enjoys, at the cost of some efficiency.

A further subtlety arises because the selected set is random. The FLCI at bound M is valid for a given selected set only if that set's residual differential trend lies in $\Delta^{SD}(M)$; since the set is random, its residual curvature is a random variable, and unconditional coverage requires M to dominate that variable with high probability.

Lemma 1 (Calibration to the random selected set)**.** *Under data-driven selection, choosing M equal to the* mean *residual curvature of the selected aggregate undercovers; unconditional coverage requires M to exceed the mean and to approach an upper quantile of the residual-curvature distribution, with the FLCI's sampling slack providing a partial buffer.*

Lemma 1 is confirmed sharply in Section 4 and is a practical warning: the sensitivity bound must be set conservatively relative to the *typical* selected set, not merely to satisfy it on average.

Finally, we caution against one otherwise-natural choice of Δ. The relative-magnitudes restriction $\Delta^{RM}(\bar{M})$ of Rambachan and Roth (2023), following Manski and Pepper (2018), bounds post-treatment violations by $\bar{M}$ times the observed maximal pre-treatment violation. Because it anchors its bound to the observed pre-trends, it is self-undermining precisely in the regime where our method's value is in question.

Proposition 4 (Relative magnitudes is self-undermining)**.** *The identified-set half-width under $\Delta^{RM}(\bar{M})$ is proportional to the observed pre-trend magnitude. When pre-trends are uninformative about post-treatment violations, the observed pre-trends are small while the true violation need not be, so $\Delta^{RM}(\bar{M})$ produces intervals whose width tends to zero even as the bias does not, yielding* *undercoverage. A structural restriction such as $\Delta^{SD}(M)$ does not share this defect, since its bound keys off smoothness rather than pre-trend magnitude.*

3.4 Scope: when to prefer the LATT

The method's advantage over the ATT is not uniform; it is governed by how informative pre-trends are about post-treatment violations. When they are informative, selecting on them isolates credible cohorts and the LATT recovers a clean effect while the ATT estimator remains biased. When they are uninformative, selection is uninformative about the post-period, the residual violation of the selected set is as large as for any set, and the method offers no advantage over the ATT—there is no free lunch, and the identifying assumption must be defended from economic context. We regard a formal decision-theoretic characterization of the region in which the point-identified LATT dominates the set-valued ATT, in the sense of Armstrong and Kolesár (2018), as the principal open theoretical question; the simulations below map its empirical shadow.

4 Simulation study

4.1 Design

We study the method in two environments. In the *reduced-form* model we draw event-study coefficients directly from their asymptotic distribution, $\hat{\beta} \sim \mathcal{N}(\tau + \delta, \Sigma)$ with Σ known, which isolates the identification and inference mechanics in the setting where the underlying theory is exact. In the *panel* model we generate individual outcomes for a staggered design and estimate both the coefficients and their covariance from the data, which confirms that the conclusions survive realistic estimation. Unless stated otherwise, results are averages over eight to twenty thousand replications.

The cohort structure is the same throughout. There are twelve cohorts, of which some are *clean*, with $\delta_g = 0$, and the remainder *confounded*, with a nonzero differential trend. Every cohort is given the same true treatment effect, normalized to one. This normalization is deliberate: it makes the true ATT and the true LATT both equal to one, so that any departure of an estimator from one is attributable to a violation of parallel trends rather than to treatment-effect heterogeneity. It therefore isolates the identification problem that the paper is about.

A confounded cohort is parametrized by the size of its post-treatment violation, $V > 0$, and by how visible that violation is beforehand: its pre-treatment violation is $\textit{info} \times V$. The scalar $\textit{info} \geq 0$ is the master axis of the study. When $\textit{info} = 0$ the confound is entirely invisible in the pre-period, so no pre-trend-based rule can detect it; as *info* grows the confound announces itself before treatment and becomes detectable. This single parameter operationalizes the informativeness of pre-trends discussed in Section 3.

The selection rule is deliberately simple and data-driven: cohort g is retained if its estimated pre-trend is small in magnitude, $\max_e |\hat{\beta}_{g,\mathrm{pre}}(e)| \leq c$ for a threshold c. The estimator of the ATT averages the post-treatment coefficients over *all* cohorts, as a heterogeneity-robust estimator would;

Table 1: The estimand demonstration (reduced-form model). True effect = 1 for all cohorts.

	ATT estimator	LATT (ours)
Point estimate	1.267	1.013
Bias vs. truth	+0.267	+0.013
Bias as sampling noise s falls:		
$s = 0.40$	+0.269	+0.048
$s = 0.20$	+0.267	+0.005
$s = 0.12$	+0.267	+0.000
$s = 0.03$	+0.267	−0.000

the LATT averages them over the *selected* cohorts only. Because the sampling errors of a cohort's pre- and post-treatment coefficients are correlated, the selection is genuinely data-dependent in the sense of Section 3, and the pre-test bias of Proposition 3 is operative.

For the inference exercises of Sections 4.4–4.5 we use a fuller event study with four pre-treatment and four post-treatment periods and a reference period normalized to zero, and a differential trend that is a smooth curve, $\delta_e = (C/2)e^2$, whose maximal second difference is exactly C. The parameter C therefore measures the curvature of the confound and is directly comparable to the smoothness bound M of the restriction $\Delta^{SD}(M)$, which is what makes the validity statement "covers if and only if $M \geq C$" checkable.

4.2 The estimand: recovering a clean effect where the ATT is biased

We first illustrate Proposition 1 in the simplest configuration, with a subset of cohorts violating parallel trends in the post-period. Table 1 reports the result. The estimator of the ATT is biased by +0.267: it averages over all cohorts, so the offending cohorts' differential trends enter it directly. The LATT lands at 1.013, essentially on the truth, because the selection removes those cohorts from the average and, by Proposition 1, their violations then do not enter the estimand at all.

The lower panel of the table sweeps the sampling noise s, and is the more informative comparison. As precision improves the LATT's bias falls monotonically from +0.048 to zero: this residual is the pre-test bias of Proposition 3, which arises because the selection is made on the same noisy pre-trends used for estimation, and it vanishes as the selection rule becomes reliable. The ATT estimator's bias, by contrast, is frozen at +0.267 regardless of precision. The contrast is the paper's basic point in miniature: the LATT faces a noise problem, which more data cures, whereas the ATT estimator faces a wrong-target problem, which no amount of data cures.

4.3 The scope condition

Figure 1 traces the method across the informativeness axis *info* described in Section 4.1, and its four panels should be read together.

The upper-left panel plots the bias of the two point estimators against the truth. The bias of the ATT estimator is a flat line at +0.30: it never consults pre-trends, so its performance cannot depend on how informative they are. The LATT's bias begins on top of that line at $\mathit{info} = 0$ and falls to zero as pre-trends become informative. The vertical gap between the two curves is therefore exactly the method's advantage, and reading it off from left to right gives the scope condition: the advantage is nil when pre-trends carry no information about post-treatment violations, and maximal when they carry a great deal. That the two curves coincide at the left edge is not a defect but an honest reflection of the fact, argued in Section 3, that selecting on uninformative pre-trends buys nothing.

The upper-right panel explains *why* the LATT's bias falls, by plotting the identification gap of the selected set—the average post-treatment violation among the cohorts that survive selection. It tracks the LATT's bias almost exactly, falling from 0.30 to essentially zero. This confirms that the LATT's remaining bias is not an artifact of the estimator but simply the residual violation that selection has failed to remove. The lower-right panel gives the mechanism behind both: the average number of confounded cohorts that slip through the screen falls from 5.73 of six when the confound is invisible to 0.04 of six when it is plainly visible.

The lower-left panel turns to inference and reports the coverage of a point-based confidence interval for the causal target, computed both in the naive way and by sample-splitting. Both collapse at low informativeness and recover only at high informativeness. The comparison is diagnostic: sample-splitting removes any contamination from data-dependent selection by construction, so the fact that it collapses just as badly establishes that the failure is *not* selection distortion but the identification gap of the upper-right panel—the surviving cohorts genuinely still violate parallel trends. This is what motivates the sensitivity bounds of the next subsection, since no purely inferential fix can repair a violation of the identifying assumption. (At the right edge the naive interval slightly outperforms the split one, simply because splitting discards half the data once selection has become nearly deterministic and there is no contamination left to correct.)

4.4 Honest inference restores coverage

We now validate the smoothness-class FLCI of Section 3 applied to the selected-cohort aggregate, using the curved differential trend of Section 4.1 whose curvature C is directly comparable to the assumed bound M.

Table 2 is the validity check. Reading down the rows, when the assumed bound equals the true curvature ($C = M$) the FLCI covers at 94.7% and 95.2%, that is, at its nominal level; when the bound is generous ($M > C$) it covers conservatively at 100%; and when the bound is violated ($M < C$) coverage falls to zero. This is precisely the behavior an honest sensitivity interval should display—it is valid exactly on the range of violations it claims to accommodate, and it fails visibly outside it, rather than failing silently. The final column records that both point-based intervals, the naive one and the one that linearly extrapolates the pre-trend, collapse to zero coverage the instant any curvature is present, because each is centered on a biased estimate with a width reflecting sampling noise alone.

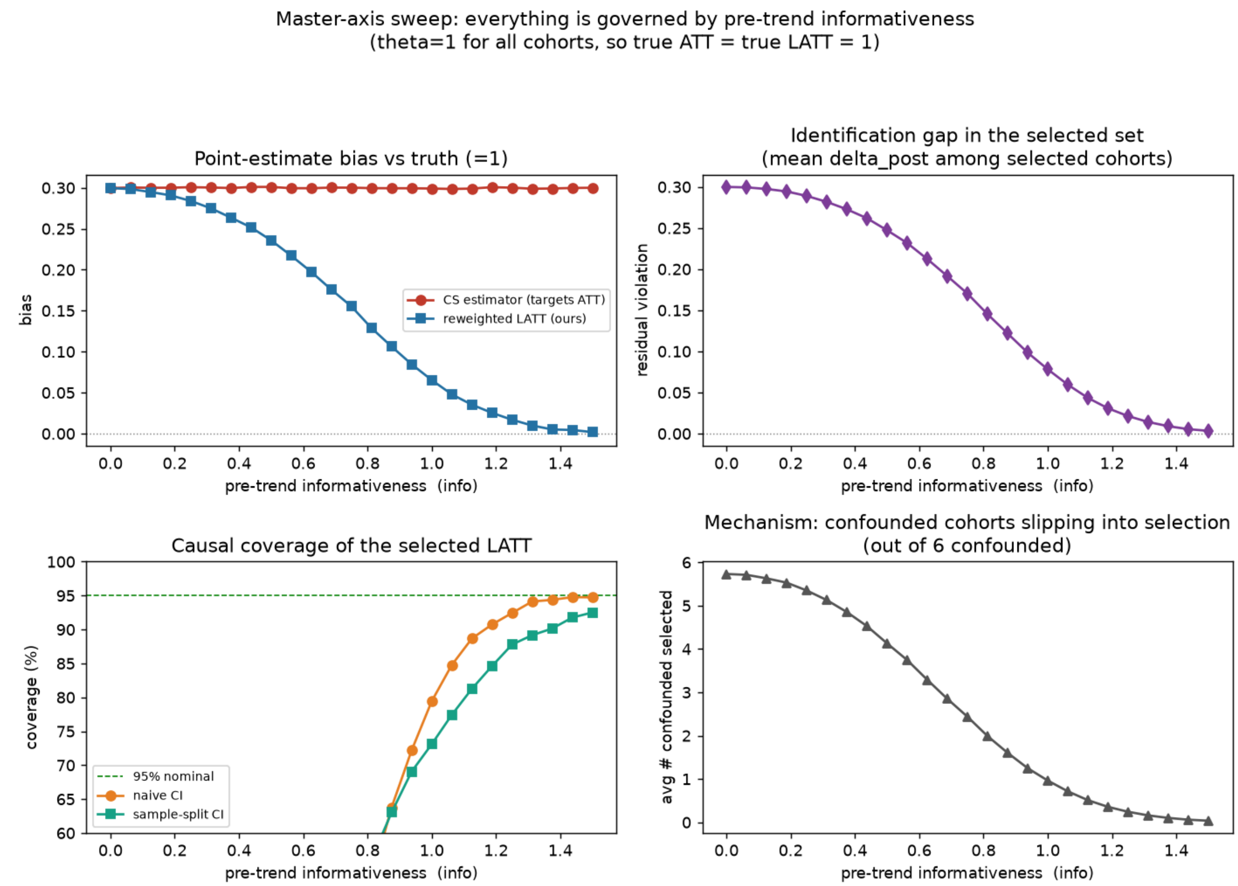

Figure 1: Scope condition. Everything is governed by pre-trend informativeness. The ATT estimator's bias is constant; the LATT's advantage is the gap between the lines. Causal-coverage collapse at low informativeness is an identification gap (surviving cohorts still violate parallel trends), not selection distortion—sample-splitting collapses identically.

Figure 2 repeats the exercise across a continuum of curvature and adds the practical output. In the left panel the two point-based intervals fall away rapidly as the confound curves, while each FLCI holds its nominal level until the true curvature reaches its own assumed bound—marked by the vertical dotted lines at $M = 0.10$ and $M = 0.20$—and declines thereafter. The middle panel records the price: interval half-width is governed by the assumed M rather than by the realized curvature, which is why the curves are flat in C and why the more permissive bound is roughly 75% wider. Honesty about a wider class of possible violations is paid for in width, immediately and proportionately.

The right panel is what a practitioner would actually report. Holding the data fixed at a curvature of 0.10, it traces the confidence band as the assumed bound is relaxed and identifies the breakdown value $M^* \approx 0.077$ at which the band first admits the truth. Reported alongside an estimate, this quantity answers the question a reader of a DiD paper wants answered: how much departure from a linear differential trend must one be willing to countenance before the

Table 2: FLCI coverage of the causal target (reduced-form model): valid iff $M \geq C$.

True curvature C	Assumed M	FLCI coverage	Point-estimate coverage
0.5	0.5 ($= C$)	94.7%	0.0%
0.5	1.0 ($> C$)	100%	0.0%
1.0	0.5 ($< C$)	0.0%	0.0%
1.0	1.0 ($= C$)	95.2%	0.0%

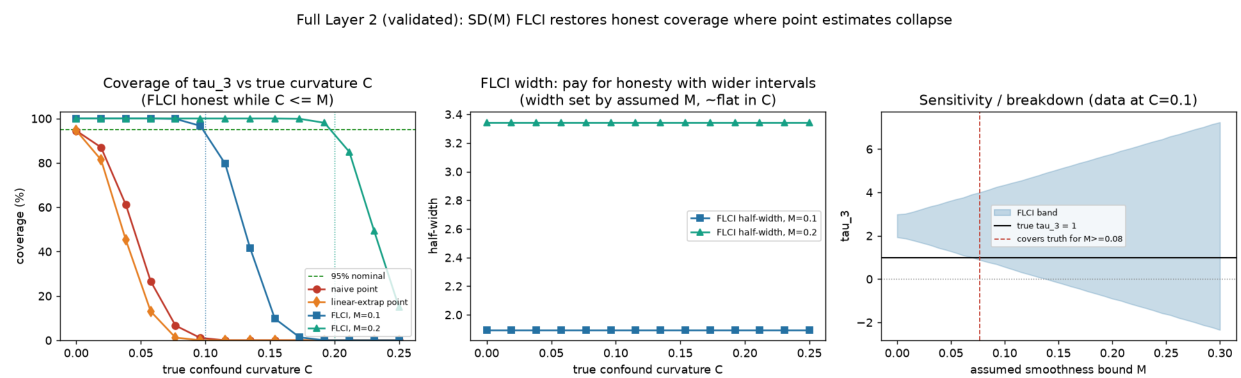

Figure 2: Sensitivity bounds restore honest coverage. Point estimates collapse as the confound curves; the FLCI is honest exactly while $M \geq C$; interval width is the price of honesty, set by M; the right panel is the sensitivity/breakdown curve.

stated conclusion no longer follows. Because $\Delta^{SD}(M)$ keys off smoothness rather than pre-trend magnitude, this exercise remains informative even when pre-trends are uninformative—in direct contrast to the relative-magnitudes restriction of Proposition 4, whose bounds would shrink toward zero in exactly that case and deliver false precision.

4.5 Selection does not distort the bounds, but calibration must respect randomness

Two questions remain for the data-driven regime. Does the fact that the selected set was chosen using the same data distort the sensitivity interval; and how should the bound M be chosen given that the selected set, and hence its residual violation, is random. Figure 3 answers both, and its two panels are best read in reverse order.

The right panel displays the distribution of the residual curvature of the selected aggregate across replications. It is visibly spread and multi-modal, with mass at roughly 0, 0.02, 0.033, 0.043 and 0.05; each mode corresponds to a different number of borderline cohorts surviving the screen in that replication. This is the concrete content of Lemma 1: the object that a fixed bound M must dominate is not a constant but a random variable, and a bound chosen to sit at its mean will be exceeded in a substantial share of samples.

The left panel confirms the consequence and settles the distortion question. Coverage is plotted against M for the full-data procedure and for sample-splitting. The two curves lie almost on top

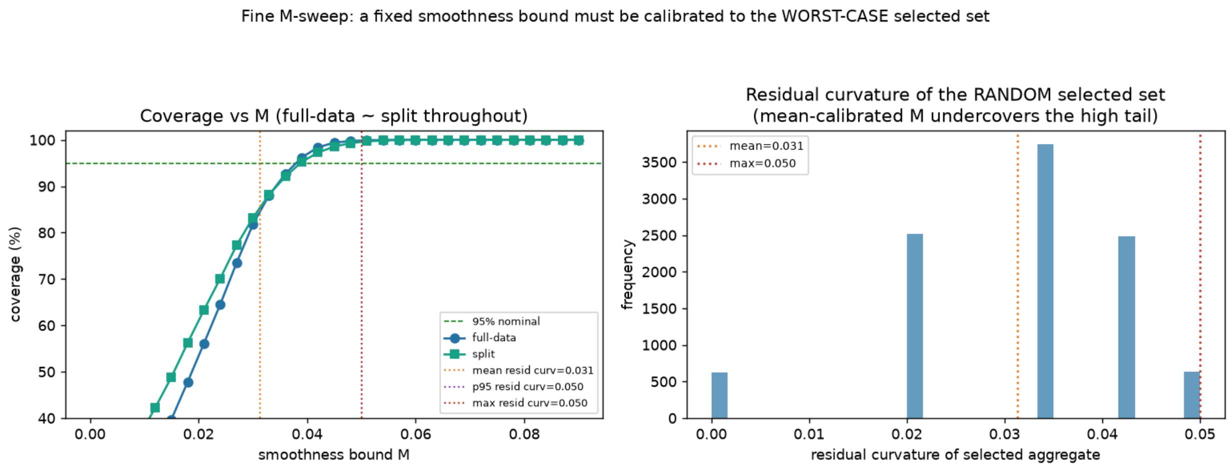

Figure 3: Calibrating the sensitivity bound to the random selected set. Full-data and split coverage cross nominal at the same M (no selection distortion); a mean-calibrated M undercovers, with nominal coverage reached above the mean residual curvature.

of one another and cross the nominal level at the same bound, $M \approx 0.039$, which establishes that data-dependent selection introduces no systematic distortion of the FLCI here; splitting buys nothing on this account, consistent with the pointwise—as opposed to worst-case—reading of Leeb and Pötscher (2005) discussed in Section 3. Turning to calibration, the vertical reference lines mark the mean, the ninety-fifth percentile and the maximum of the residual-curvature distribution. Setting M at the mean, 0.031, yields only about 88% coverage; nominal coverage requires $M \approx 0.039$, comfortably above the mean though still below the strict maximum of 0.050, the shortfall being absorbed by the sampling slack already built into the interval. The practical recommendation follows directly: calibrate the sensitivity bound to the residual violation of a *typical* selected set, not to its average, and treat the resulting bound as a floor rather than a target.

4.6 Panel confirmation

The exercises above all draw coefficients from their asymptotic distribution with a known covariance, which is the right setting in which to check identification and inference mechanics but is not the setting a practitioner faces. We therefore repeat the estimand demonstration on generated panel data. Individual outcomes are simulated for a staggered adoption design with unit and period effects and idiosyncratic noise; cohort-level pre- and post-treatment coefficients are then estimated from genuine difference-in-differences contrasts against a common never-treated comparison group; and the covariance matrix is estimated from the data rather than supplied.

The conclusions are unchanged. Across the informativeness axis the ATT estimator's bias remains flat near +0.30 while the LATT's bias falls toward zero, reproducing the upper-left panel of Figure 1 with estimated rather than known quantities. The estimated covariance is well calibrated in the region where selection is effectively deterministic, with coverage of 94 to 95% at high informativeness, which confirms that the inference machinery is not being flattered by the reduced-form assumption of a known Σ.

One discrepancy is worth reporting because it is informative rather than incidental. At intermediate informativeness, where selection is genuinely stochastic, the estimated standard error of the LATT (0.079) falls short of its Monte Carlo counterpart (0.119). The gap is the component of variance induced by the randomness of the selection itself, which a standard error computed conditional on the realized selected set does not capture. This is an independent confirmation, from the panel side, of the inferential issues discussed in Section 3: conditioning on a data-dependent selection understates uncertainty, which is precisely why the honest object to report is the sensitivity interval of Section 4.4 rather than a conventional confidence interval around the point estimate.

5 Conclusion

We have argued that when parallel trends fails for some treated cohorts, a productive response is to change the estimand rather than to defend or bound the ATT. The credible-subpopulation LATT is point-identified precisely when the ATT is not, is estimated by a transparent reweighting of standard group-time effects, and can be accompanied by honest sensitivity bounds on the residual violation of the selected set. The approach is a specific instance of a general principle—retreat to a credibly-identified subpopulation—already established in the analysis of instrumental variables and of limited overlap.

We have been explicit about the method's advantages and its weaknesses. Its advantage over the ATT is that it delivers a sharp, credibly-identified answer where the ATT estimator is biased and the ATT sensitivity set is wide; its transparency about which comparisons identify the effect is inherited from Callaway and Sant'Anna (2021). Its weaknesses are equally concrete. The estimand is a subpopulation effect and must be interpreted as such, with the composition of the credible set described rather than assumed away. Its advantage is contingent on pre-trends being informative about post-treatment violations, and when they are not, only economic context can carry the identifying assumption. Under data-driven selection the estimator carries a vanishing pre-test bias, and a worst-case uniform guarantee is unavailable without ex-ante selection, sample-splitting, or randomization; what we can offer is pointwise validity away from the selection boundary, together with simulation evidence that the distortion is small in plausible configurations and arises where theory predicts. The sensitivity bound must be calibrated to the typical selected set, and the relative-magnitudes restriction should be avoided in favor of a structural one, since the former is self-undermining under uninformative pre-trends.

The principal open question is theoretical: a decision-theoretic characterization of the region in which the point-identified LATT dominates the set-valued ATT would convert the empirical scope condition documented here into a formal recommendation, and would sharpen the choice between the two responses to imperfect parallel trends. We regard the framing developed here—estimand choice as the lever, with credibility purchased by narrowing the target—as the paper's contribution, and the accompanying honesty about scope as inseparable from it.

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